

Grapes

ONE

The Berry

A grape is a berry. Botanists mean something exact by that word: a fleshy fruit that grows from a single flower with a single ovary, its seeds buried in the pulp. By this rule the tomato is a berry, and so are the banana and the eggplant, while the strawberry and the raspberry are not.^a¹ The grape qualifies without argument.

Nearly all the grapes we eat, dry, and press come from one species, *Vitis vinifera*, the wine-bearing vine.^{*}² The rest of the genus, several dozen species, grows wild across the temperate Northern Hemisphere, and a good number of them are American. For most of history this mattered to no one in Europe.

^{*} *Vitis* is the Latin for vine.

¹The strawberry is an accessory fruit. Most of its flesh grows from the receptacle, the part of the flower that holds the ovaries, and the specks on the outside are the true fruits, each one an achene.^a The raspberry is an aggregate: a cluster of many small fruits, each from a separate ovary of one flower. The strawberry comes back later in the chapter, e, p. 2.

²*Vinifera* is Latin for wine-bearing, from *vinum*, wine, and *ferre*, to bear. The English word *grape* took a stranger road. It comes from Old French *grape*, a bunch of grapes, and probably before that from a Germanic word for a hook.^b So the name of the fruit first meant the tool, then the cluster, and only last the single berry.

^aAn achene is a small dry fruit with one seed that does not split open when ripe. A sunflower seed in its striped shell is also an achene, although botanists give the sunflower's version its own name, cypsela.

^bPresumably the hook used to cut the bunches. The same root is thought to lie behind *grapple* and French *grappin*.

Then, in the 1860s, vines in the south of France began to die.³ Over the next decades the blight moved through France and into the rest of Europe. The cure, found after much argument, was to graft European vines onto American roots. Most of the world's wine grapes still grow that way, a European plant standing on American feet.⁴

The Americans, meanwhile, had grapes of their own. In the 1840s Ephraim Wall Bull, in Concord, Massachusetts,^{*} began planting seeds of a wild native species, *Vitis labrusca*, and selecting among the seedlings. The Concord grape came out of that work.⁶

In the Tokaj hills of north-eastern Hungary,[†] the growers wait for a mould. Late in a good autumn, a fungus pierces the skins of the ripe berries and the water inside evaporates. What remains shrivels into something between a raisin and a jewel, and it makes one of the great sweet wines.⁷

^{*} The same Concord as Emerson and Thoreau.⁵

[†] Older English spells it Tokay.

³The cause was an insect smaller than a pinhead, now called *Daktulosphaira vitifoliae* but known to everyone by its older name, phylloxera.^a It had come from North America, probably on vines brought over by collectors. American vines had lived with it for a long time and tolerated it. European vines had not.

⁴The exceptions are places the insect never reached or cannot live, such as vineyards on pure sand.^b Chile is the example usually given.

⁵Thoreau was born there, in 1817.

⁶Its flavour, which Europeans tend to call foxy, is the flavour most Americans know as grape: the flavour of grape jelly and grape soda. In 1869 Thomas Welch, a dentist in Vineland, New Jersey, heated Concord juice to stop it from fermenting, and so made an unfermented wine for church communion. For the town, see ^{*}.^c

⁷The fungus is *Botrytis cinerea*, noble rot. The shrivelled berries are called *aszú*.^d In the wrong weather the same fungus is simply grey mould, the grey fur on a forgotten punnet of strawberries.^e

^aFrom Greek *phyllon*, leaf, and *xeros*, dry. The name describes the damage the insect does to leaves, although the fatal damage is to the roots.

^bSand seems to hinder the insect's movement through the soil. Why, exactly, is a question I have not yet looked into.

^cWelch was a supporter of temperance. He was applying a method that Louis Pasteur had worked out only a few years earlier, in the mid-1860s.

^dThe word is related to the Hungarian verb for drying up. For the spelling of the place, see [†].

^eHere the strawberry returns (see 1, p. 1), as it will throughout this book, and with it the question of what counts as a berry, , p. 1.